# Slide 1: Recommendation - BLUF

Approve KES 14.2M safety-stock and distribution reallocation by 30 September 2026.

Action: Approve KES 14.2M Reallocation | Target Deadline: 30 September 2026

Key Recommended Initiatives:
1. Increase safety stock by 18% at five high-lead-time retail stations.
2. Rebalance depot-to-station flows using PuLP-optimized linear programming.
3. Add two dispatch coordinators for holiday and peak-season truck scheduling.

Expected Impact:
• Customer fill rate improves from 96.5% to 98.5%.
• Stockout-driven margin loss falls by 42%.
• Annualized protected margin exposure: KES 46.8M (Net Benefit: KES 32.6M).

# Slide 2: Strategic Investment Context

Macro Factors & Network Scope:
• 8 regional stations supplied via Nairobi Depot (380k L capacity) and Mombasa Depot (340k L capacity).
• Retail demand growth of +12.8% YoY driven by infrastructure expansion.

The Strategic Problem:
• Fixed static reorder points fail to adjust for holiday spikes and supplier delivery variability.
• Emergency expedited freight expenses reached KES 12.4M annually.
• Customer service level agreements (SLAs) incurred KES 4.8M in penalties.

Strategic Alignment: Transitioning from reactive firefighting to predictive inventory optimization.

# Slide 3: Safety & Compliance Performance (Week 7 Recap)

Recap of Safety Metrics & Environmental Analytics:

- Total Recordable Incident Frequency Rate (TRIFR): Reduced from 2.4 to 1.1 per million hours.
- Regulatory Compliance Rate: 98.2% across fuel depot storage and transport operations.
- Environmental Spill Risk: Zero major environmental containment breaches in Q2/Q3.
- Standardized Protocol: Implemented automated safety alert escalation for depot loading racks.

Safety Rationale: Operational stability and zero-harm logistics are prerequisites for supply chain optimization.

# Slide 4: Equipment Health & Predictive Maintenance (Week 6 Recap)

Recap of Predictive Maintenance & Reliability Analysis:
- Remaining Useful Life (RUL) Models: Deployed sensor-based predictive maintenance on primary depot pumps.
- Weibull Reliability Analysis: Identified pump impeller wear patterns, preventing catastrophic outages.
- Unplanned Downtime Reduction: Depot pump failure risk reduced by 64%.
- Operational Synergy: High pump reliability ensures distribution plans can be executed without fleet bottlenecks.

# Slide 5: Supply Chain Performance - Prophet Demand Forecast

Demand Forecasting Methodology & Results:
• Dataset: 365 days of historical daily litres sold across 8 retail stations.
• Model Architecture: Prophet time-series model incorporating annual, weekly, and custom holiday regressors (Easter, Madaraka, Mashujaa, Jamhuri, Christmas).
• Forecast Accuracy: Achieved 4.12% MAPE and 14,250 L RMSE, outperforming 30-Day Moving Average (8.65% MAPE).
• Insights: Holiday surges increase peak demand by +45%, requiring pre-positioning 7 days prior.

# Slide 6: Inventory Optimization & Stockout Reduction

Safety Stock & Reorder Point (ROP) Policy Simulation:
• Safety Stock Formula: SS = Z * sqrt(L * s_d^2 + d^2 * s_L^2) for 98.5% Service Level (Z = 2.17).
• Coverage Gap: 5 of 8 stations currently operate below required inventory coverage days.
• Policy Simulation Results:
- Policy A (Without SS): Stockout frequency = 14.2% of operational days.
- Policy B (With SS): Stockout frequency falls to < 1.2%, protecting customer fill rate.
• Action Rule: Automated reorder triggers when station inventory drops below calculated ROP.

# Slide 7: Linear Programming Distribution Optimization

Depot-to-Station Transportation Cost Minimization (PuLP LP Model):
• Objective: Minimize total transit cost sum(c_ij * x_ij) subject to depot capacity & station reorder demand.
• Optimized Allocation Summary:
- Nairobi Depot supplies Central, Nakuru, Thika, Nyeri, Machakos (Lowest transit cost per litre).
- Mombasa Depot supplies Mombasa Hub, Kisumu Port, Eldoret Depot.
• Financial Efficiency: Reduced total fleet distribution cost by 11.4%, saving KES 3.8M quarterly.

# Slide 8: Strategic Recommendations & CFO Q&A; Briefing

Three Actionable Recommendations:
1. Capital Reallocation: Reallocate KES 14.2M working capital to high-lead-time station safety stock by 30 Sept.
2. Dynamic ROP Integration: Connect Prophet 60-day forecast outputs directly to ERP replenishment triggers.
3. Logistics Coordination: Hire 2 dispatch coordinators for peak-season fleet routing.

CFO Q&A; Simulation:
Q: 'Why hold KES 14.2M in safety stock instead of relying on expedited freight?'
A: Holding targeted safety stock costs ~KES 1.8M in annual carrying costs, whereas stockouts & emergency freight cost KES 46.8M. Safety stock provides a 2.3x net ROI with payback in <6 months.